

Kolagani Pavan Kalyan

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EXPERIENCE

Founding Engineer

Nov 2025 – Aug 2026

Cortex Technology (cor-tex.solutions), voice agents for restaurants and clinics

Remote

- Callers were being cut off mid sentence, because one turn-detection profile was serving both the greeting and the conversation. Split it in two, a 0.5 threshold with 450 ms of trailing silence once the call is running and a stricter 0.8 with 700 ms during the greeting, so line noise on pickup stopped registering as speech and the agent held its turn until someone actually spoke.
- The agent sounded like it was calling from an empty room, which callers read as a robot. Built an ambience mixer that blends a restaurant bed into the outbound stream at 20 ms mu-law frames with crossfade, level-clamped so it never masks the speech underneath. Also cut model latency on live calls by moving to smaller speech models, then reverted when word accuracy dropped.

Technical Partner and Shareholder

Jul 2026 – Present

Dhanam Collections (dhanamcollections.com), handloom retail

Andhra Pradesh

- Hosting was climbing on 12.4M serverless invocations a month for a shop with 45 products. Traced it to link prefetch: every product tile entering the viewport was server-rendering a page nobody opened. Disabled prefetch on the dynamic routes and the bill fell 72%, \$87 a month to \$25.
- Only engineer on the system: 285 commits in 43 days, 24,676 lines of TypeScript, 43 page routes, 38 API routes, 20 Postgres tables, live card payments. INR 5 lakh in sales across 200+ orders to date, on 32,376 visitors and 299,734 page views in the first week at a 12% bounce rate, with function errors under 0.1% and no timeouts through the order spikes.

Co-Founder

Jun 2026 – Sep 2026

OnePurplePen (onepurplepen.com), building OneDB

Remote

- Built the directory engine behind OneDB: per-item slug pages, an MCP directory, per-directory titles and descriptions, and icon upload, so a new collection could be listed without shipping a deploy for it.
- Search crawlers could read the listings but agents could not. Wrote the machine-readable surface, `11ms.txt` plus JSON-LD structured data on every item, so an LLM crawler resolves a listing to structured fields instead of scraping the rendered page.

OPEN SOURCE

LMCache / ai-dynamo/dynamo (NVIDIA)

Python, Rust | 11.6k / 8.0k stars

KV cache for LLM serving, inference runtime

Sep 2026

- LMCache: two different tag sets serialise to the same cache key. That key is both the on-disk filename and the remote object key, so one cached entry silently reads back as another (#4904).
- dynamo: request-trace header capture matched only exact keys, so prefixed headers were missed and sensitive ones could not be excluded from traces. Added prefix matching and a denylist, +388/−37 across 6 files (#14329).

supermemory

TypeScript | 29.2k stars | memory API for LLM apps

2 merged, 1 issue filed

Jun 2026 – Aug 2026

- A retrieval against an empty profile returned an error rather than an empty result, so every first-time user looked like a failure instead of a cold start (#1164).
- The OpenAI SDK stopped importing on a fresh install after an upstream 3.x release. Filed it (#1235), then sent the version pin that fixed it (#1236).

tinygrad / HelixDB

Python, Rust | 33.5k / 5.9k stars

autodiff, vector database

Apr 2026 – Sep 2026

- tinygrad: `TRUNC` had no backward pass, so any gradient flowing through it was dropped silently rather than erroring (#15925, merged).
- helix-db: the NEON and scalar vector-distance kernels disagree on 38% of random pairs at 22 dimensions and 93% at 1536. They sit under similarity search, so the two return different neighbours for the same query (#1075).

PROJECTS

ciot | C++23, SIMD | 23 stars

- CPU inference for ternary neural nets. Weights are −1, 0 or 1, so the matmul is adds and subtracts and no multiplies at all. One binary, no runtime to install.

strata | Rust

- Inference engine for a 200B+ mixture-of-experts model on a 16 GB laptop, where NVMe sits on the critical path of every token rather than only in cold start.

helix | Python, PyTorch

- A sequence memory cell that holds state as accumulating phase angles instead of decaying activations. One neuron, 169 parameters, learns 16-bit parity to 100%, where a 128-unit GRU with 50,433 parameters sits at chance for 400 epochs: a contractive update destroys the early bits before the network sees the later ones, and phase accumulation does not. Nine benchmark scripts and the plots are in the repo, including the sine-wave regression it loses to GRU on.

TECHNICAL SKILLS

Languages: Python, TypeScript, Rust, C++23, Go

AI systems: OpenAI Realtime, ElevenLabs, streaming speech pipelines, turn detection and barge-in, KV caching, retrieval and memory layers, quantised and ternary inference, mixture-of-experts offload

Product: Twilio and Telnyx telephony, mu-law audio, Next.js, Node, PostgreSQL, serverless cost analysis

Testing: differential and property testing, numerical equivalence under rounding, evaluating a change against a live-call deadline

EDUCATION

Amrita Sai Institute of Science and Technology

B.Tech, Electronics and Communication Engineering

Paritala, India

Graduated 2026